CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

REVISED TOPIC

AOI Decision Gate

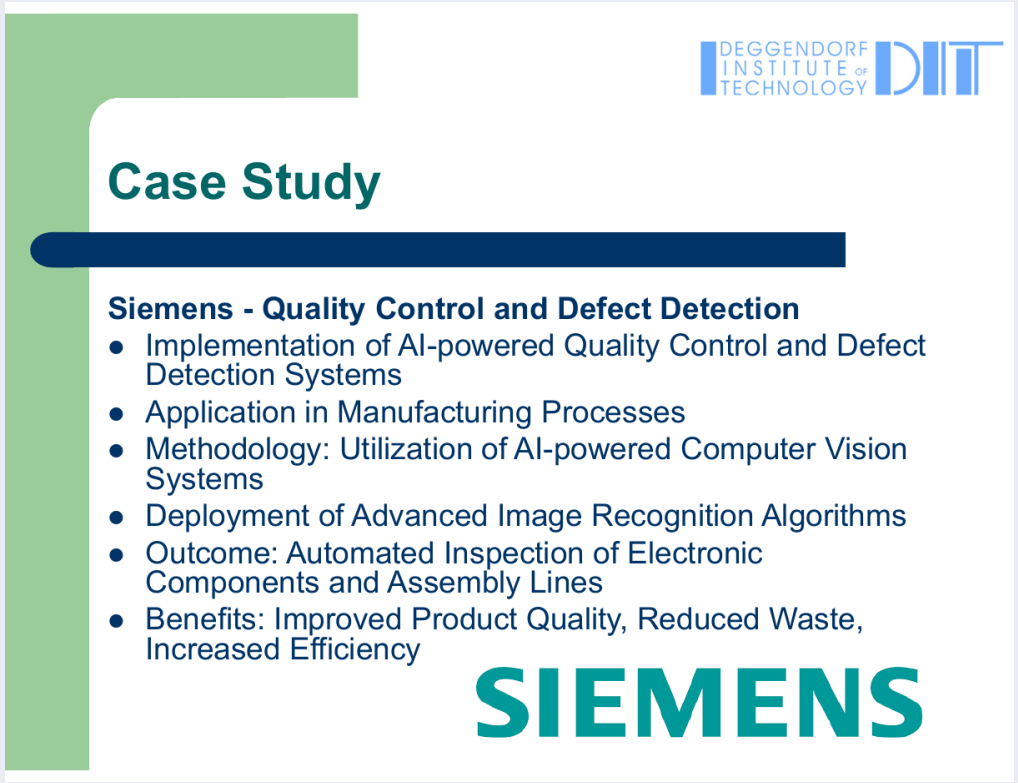
AI decision gate for false-call reduction in automated optical inspection of electronics assembly

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Team 1 · Guided by Dr. Hamidreza Heidari · 23 October 2026 · github.com/bodhi584/aoi-decision-gate

WHERE THE TOPIC COMES FROM

Based on the lecture example: Siemens, p. 18



Lecture slide	Our project
AI-powered quality control and defect detection	**AI decides: real defect or false call**
Application in manufacturing processes	**SMT electronics assembly, after soldering**
AI-powered computer vision systems	**Works on the AOI camera system's measurements**
Advanced image recognition algorithms	**Classifier on AOI image measurements**
Automated inspection of electronic components	**Real Siemens line data (Bavaria)**
Improved quality, reduced waste, efficiency	**Fewer manual checks, no extra defects released**

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THE INDUSTRIAL PROBLEM

Too many false alarms after optical inspection, today

98 %

of the boards flagged by the AOI machine are fine (false calls) in the Siemens line data.

Every one goes to a manual inspection station: “a major manual labour effort” (Pfab et al. 2024, Siemens).

21 Oct 2026 · Siemens and NVIDIA, Erlangen electronics factory

AI second check of AOI results: is it a real defect or a false call?

Project targets: about 40 % fewer false calls, reduced manual verification.

19 Oct 2026 · Siemens, Erlangen

Factories in high-wage countries need 7–9 % more productivity every year.

Siemens invests €500 million in Erlangen as a showcase for AI in manufacturing.

Sources: Siemens press release, 21 Oct 2026; technology.org, 20 Oct 2026; Pfab et al. 2024, Data in Brief 53:110110

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KEY FACTORS

What decides whether the gate helps the line

False-call rate	Each false call costs operator time at the manual station
Missed defects	A released defective board reaches the customer: must stay ≤ 1 %
Drift over time	Materials, settings and products change; last month’s model may not fit
Class imbalance	Only about 1 % of flagged joints are real defects
Label quality	Labels come from operators and may contain errors

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INPUT → SYSTEM → OUTPUT

Where the gate sits on the line

On the line today

SMT soldering

AOI camera checks every joint against fixed limits

Flags suspicious boards

INPUT

AOI data per board

~70 measurements per joint

Inspection type

Component type

AOI error code

SYSTEM (OURS)

AI decision gate

Classifier trained on past manual-inspection results

Release threshold: at most 1 % of defective boards released

OUTPUT

Release

likely false call → next step

Manual inspection

possible defect → operator checks

Today

Every flagged board is checked by hand, and 98 % of them are fine.

With the gate (goal)

Fewer manual checks, same quality: at most 1 % of defective boards released.

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METHODOLOGY

CRISP-DM, evaluated by a replay simulation of the line

1 · Business understanding	2 · Data understanding	3 · Data preparation	4 · Modelling	5 · Evaluation	6 · Deployment
false-call problem, KPIs, limits	structure, imbalance, drift	joints → boards, strict time order	gradient-boosted trees; neural network	replay simulation, production KPIs	replay demo, run instructions

Replay simulation (our “baseline simulation”)

The 132 recorded days are played back in time order. Each day the gate decides every flagged board with a model trained only on earlier days, exactly as on the real line.

We compare update policies (train once, retrain weekly on all past days, retrain on recent days) and measure released false calls, released defects and saved inspections.

Why not a random split? It mixes future and past and overstates what a factory gets (next slides).

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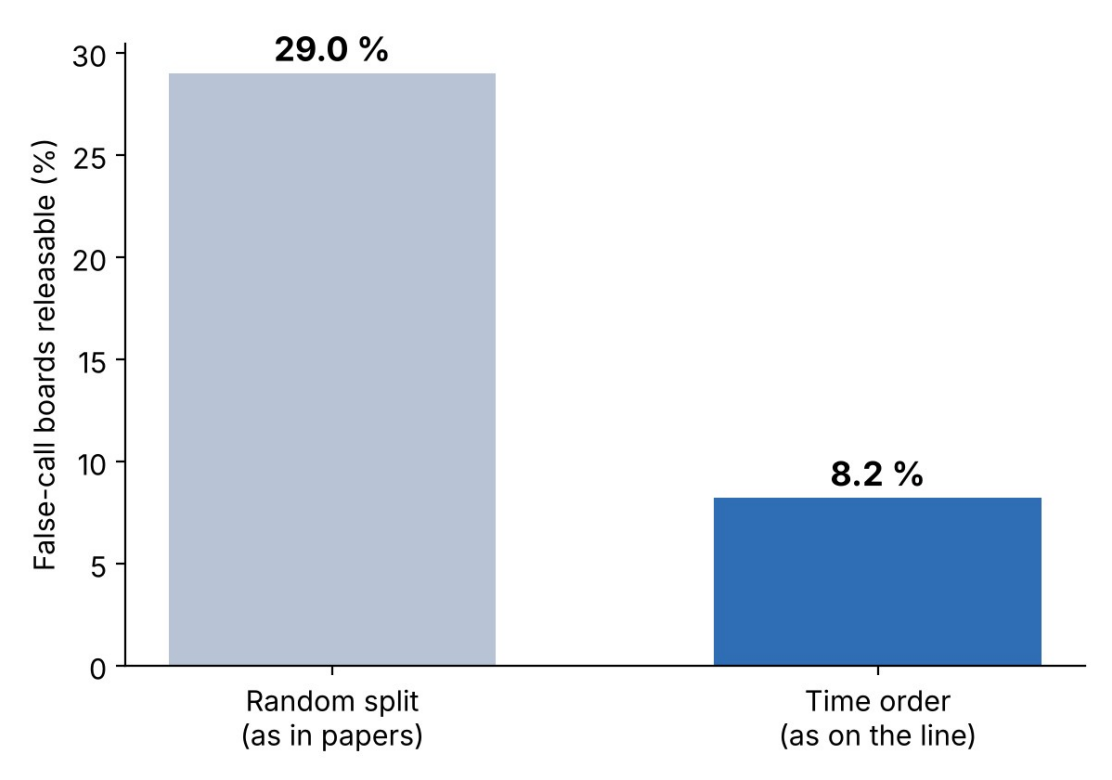
DATA

Real production data published by Siemens

Source	Pfab, Eichler, Mallandur, Rothering (2024), Siemens AG, Data in Brief 53, 110110
Line	One SMT production line of Siemens AG in Bavaria, 132 days
Size	440,274 flagged solder joints on 39,742 boards; 5 inspection types
Features	~70 AOI measurements + component type, error code, angle (anonymised, scaled 0–1)
Label	Manual inspection result: false call or true defect (875 defective boards)
Drift	Early vs. late days separable from measurements alone: AUROC 0.966
Licence	CC BY-NC 3.0 (Siemens AG): downloaded by script, not redistributed

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Time order is much harder than a random split



What it shows

Gradient-boosted trees, trained on the first 70 % of records.

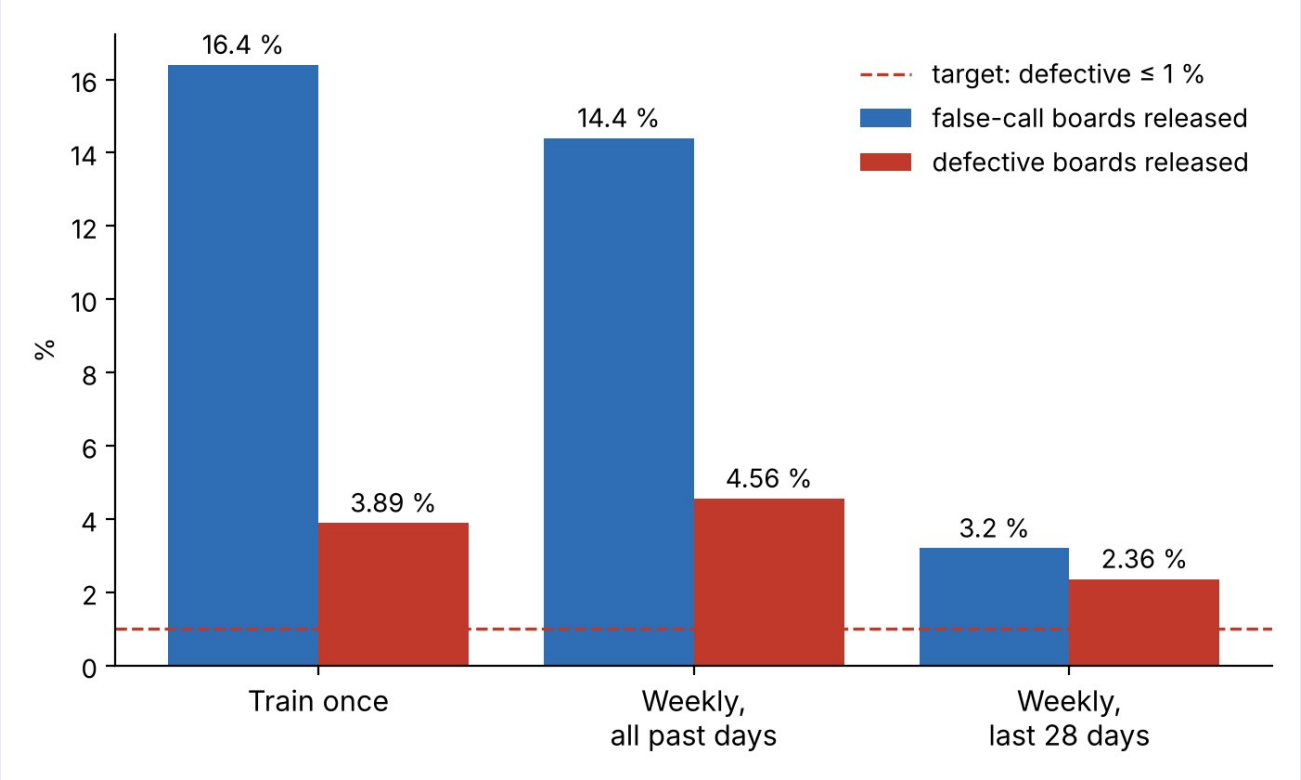
Share of false-call boards that could be released while 99 % of defective boards still go to manual inspection.

Random split: 29 %, AUROC 0.966. Time order: 8.2 %, AUROC 0.889.

The line changes over time: drift is the real challenge, and papers with random splits overstate results.

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Thresholds from past days do not hold on later days



What it shows

39 production days, 17,366 flagged boards replayed.

Up to 16 % of false-call boards released (2,750 manual inspections saved).

But 2.4–4.6 % of defective boards released: the 1 % limit is broken by drift.

→ Core of the project: drift detection and update policy.

GOALS AND KPIS

Fixed now, before the main experiments

	Baseline today	Target (12 Jan)
Defective boards released	2.4–4.6 %	**≤ 1 %**
False-call boards released (inspections saved)	8.2 %	**≥ 25 %**
Drift alarm	none	**alarm when performance drops**
Model comparison	trees only	**trees vs. neural network**

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What the project covers, and what it does not

Conditions

One Siemens SMT line, five inspection types, 132 days

Gate placed after the AOI, before manual inspection

Decisions per board; at most 1 % defective boards released

Public data only, no personal data

Limitations

Measurements anonymised: no physical names

Labels from operators may contain errors

No images; no other lines or plants

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Risk: drift limits the release rate → fallback: report what works and why

Everything on GitHub, reproducible

Tech stack

Python · pandas · scikit-learn (gradient-boosted trees)

PyTorch (neural network, from P3)

pytest + GitHub Actions: tests on every push

Neural networks (lecture p. 12, 17) · big data processing (p. 10)

No large language model

Deliverables

github.com/bodhi584/aoi-decision-gate

Decision gate + replay simulation (code, tests, CI)

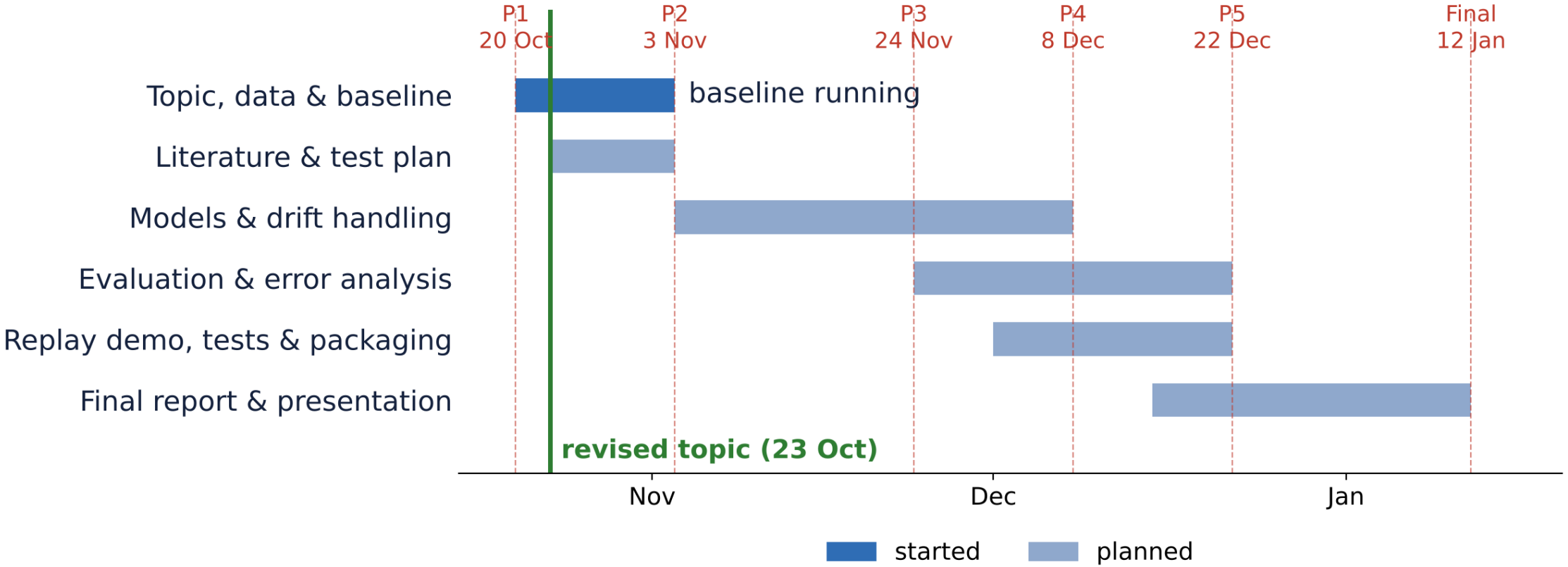
Result files for every number we report

Replay demo + demo video (P5)

Final report and presentation

TIMELINE

Milestones follow the Software & Simulation track (p. 33)



REFERENCES

Sources

Pfab, K., Eichler, R., Mallandur, A., Rothering, M. (2024). Data of automated optical inspection of surface-mounted technology electronic production. Data in Brief 53, 110110.

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Albayari, J., Patra, P., Patel, D., Santos, D. (2025). Enhancing false call detection in automated optical inspection using XGBoost classifier and XAI algorithms. FAIM 2025, Springer.

Schmitt, J., Bönig, J., Borggräfe, T., Beitinger, G., Deuse, J. (2020). Predictive model-based quality inspection using machine learning and edge cloud computing. Advanced Engineering Informatics 45, 101101.

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Lecture slides, Case Study Intelligent Systems in Production, p. 10, 12, 17, 18, 33.

Thank you

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